



Optimizing Track Maintenance Limits Using Validated Vehicle-Track Interaction Simulations

Cory Hogan – Director, Vehicle Dynamics Studies
ENSCO

B O S T O N

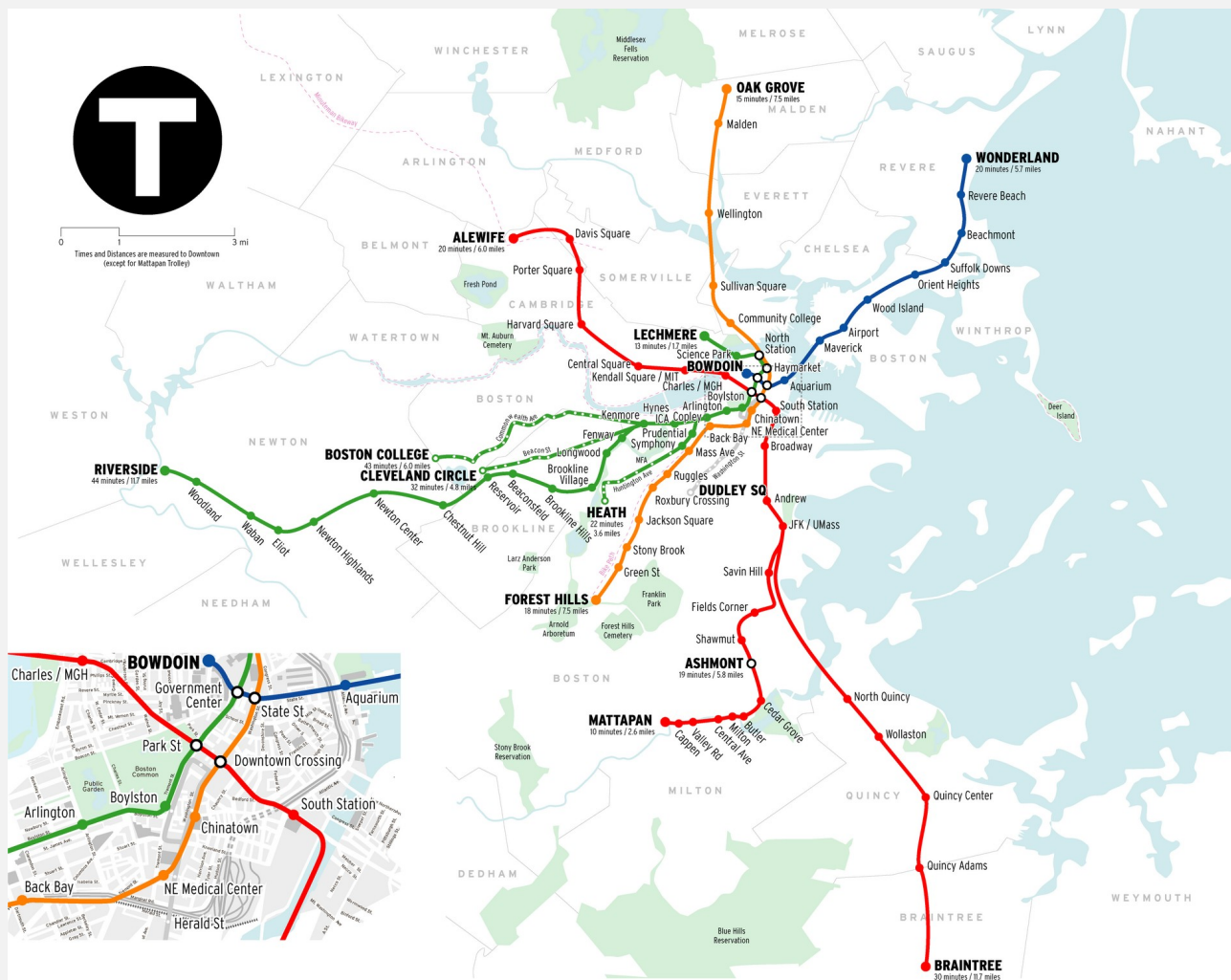


Modeling & Simulations Enable System-specific Decisions—Not Simply More Restrictions

- Simulations evaluate the interacting **vehicle, wheel, rail, track geometry, and operating conditions as a system.**
- They identify where restrictions are genuinely necessary based on the system specific characteristics—and where existing limits may be unnecessarily conservative.
- System-specific models support limits appropriate to the realities of each operation
- Better evidence can produce **more targeted, defensible, and less disruptive maintenance limits** while preserving safety.
- **The objective is not fewer safeguards—it is better-targeted safeguards supported by system-level evidence.**



Four Principal Lines — Plus the Mattapan Light-rail Shuttle



RED LINE

Heavy rail • Ashmont/Braintree branches

ORANGE LINE

Heavy rail • Oak Grove–Forest Hills

BLUE LINE

Heavy rail • third rail in tunnel / overhead outdoors

GREEN LINE

Light rail • central subway plus B, C, D and E branches

MATTAPAN LINE

2.6-mile light-rail shuttle • Ashmont–Mattapan

Each has a distinct vehicle/track environment.



Rolling Stock



RED

- CRR 1900-series fleet renewal

ORANGE

- CRR 1400-series fleet renewal

BLUE

- Siemens 0700-series; dual power collection

GREEN

- Type 7, 8 and 9 LRVs; Type 10 ahead

MATTAPAN

- Legacy PCC streetcars; Type 9 transition planned





Historical Green Line Incidents Left a Legacy of Generally Tightened Maintenance Limits

Type 8 vehicles entered limited service in 1999, but a series of derailments led to withdrawal, testing, and corrective action.

Investigations identified wheel-climb susceptibility associated with reverse curves, wide gauge, elevated angle of attack, speed, and the original wheel/flange geometry (63 Degree Flange Angle).

Responses included a 75° interim wheel profile, wheel-rail profile grinding, rail replacement, restraining-rail improvements, and adoption of shorter-chord geometry controls.

The resulting conservative maintenance thresholds increased the likelihood that measured exceptions would trigger speed restrictions, contributing to the operational burden of managing slow orders





Summary of Recent and Ongoing Network-wide Improvement Efforts



TRACK IMPROVEMENT

- Nearly **251,500 feet** of new rail installed
- More than **220 speed restrictions** removed
- Subway system became **slow-zone-free** in **December 2024** for the first time in approximately 20 years
- Approximately **2.4 million weekday passenger-minutes** saved across all lines

FY26–30 COMMONWEALTH-BACKED MBTA INVESTMENT

- **Approximately \$1.14 billion** in state-supported MBTA investment
- **\$484.7M — Red and Orange Line vehicles and infrastructure**
 - Approximately **\$422M** for vehicle investments
 - Approximately **\$62M** for associated infrastructure, including Cabot and Wellington
- **\$177.7M — power reliability, system reliability, accessibility, and resiliency**
 - Approximately **\$83M** for rail reliability, including track and power
 - Approximately **\$95M** for accessibility and resiliency
- **\$218.2M — new Commuter Rail bi-level coaches**

NEW GREEN LINE TYPE 10 VEHICLES

- **102 next-generation Type 10 vehicles** ordered from CAF USA
- Intended to replace the aging **Type 7 and Type 8** fleets

➤ **Key Question**

With billions invested in system improvements—and the Green Line shallow-flange issue now addressed—can validated, system-specific maintenance limits be established that safely preserve these improvements, minimize unnecessary operational impacts, and provide sustainable thresholds that the MBTA can maintain long-term?

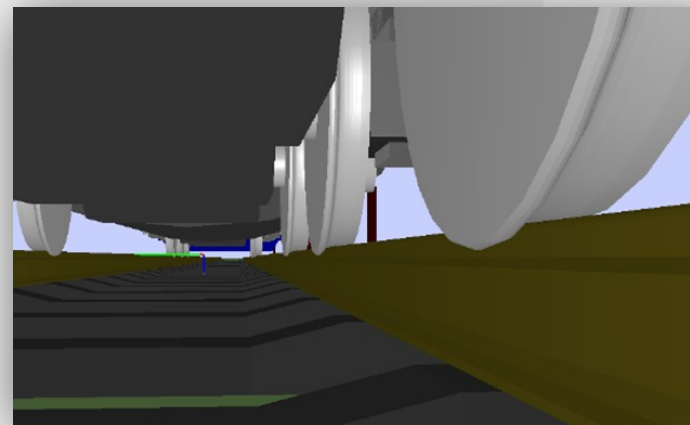
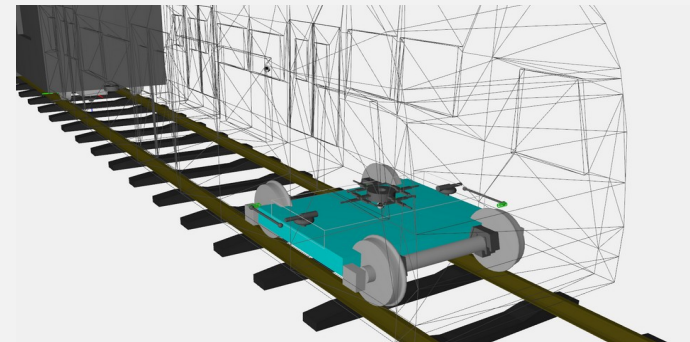
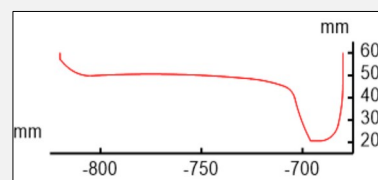
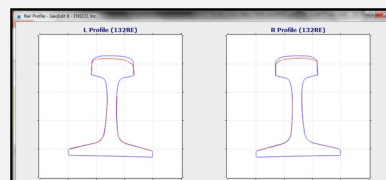
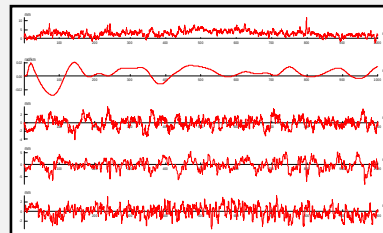


Evidence-Based Expansion of Limits

- Benchmarked practices and limits from comparable transit operations.
- Conducted instrumented testing to measure actual vehicle–track response.
- Created and validated the MBTA digital twin model using the measured results.
- Used exploratory simulations to evaluate expanded maintenance limits more appropriate to the given systems.
- Established expanded but safe system-specific limits supported and informed by industry experience, physical testing, and validated modeling.



Measurements, Simulations, and The Digital Twin



- A digital twin is a virtual model of a physical process
- The virtual and physical worlds are paired via data (i.e., track geometry measurements)
- Data analytics and simulations can predict problems, evaluate alternatives, guide decision making and optimize operations.



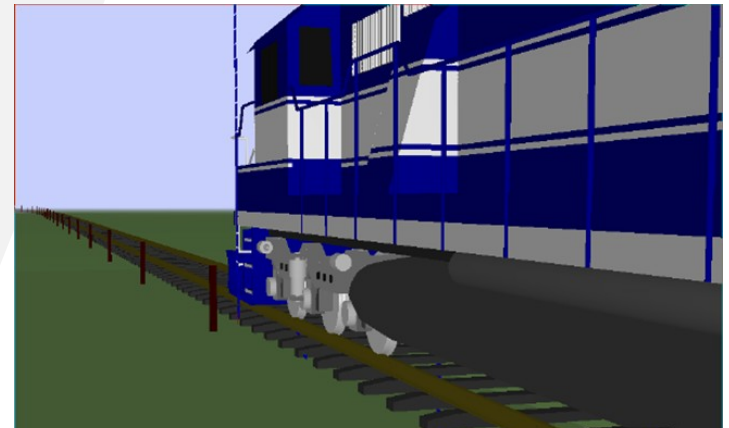
VAMPIRE® Vehicle Dynamics Modeling Software

Vehicle
analysis
Modeling
Package
In the
Railway
environment

VAMPIRE® is a multi-body rail vehicle dynamics modeling package developed by Resonate. Resonate was formed from its predecessors, Delta Rail, AEA Technology Rail and British Rail Research.

Since around the late 1970's, routines have been written to describe and predict rail vehicle behavior primarily for the purposes of design evaluation and optimization, accident investigation and safe operations setting.

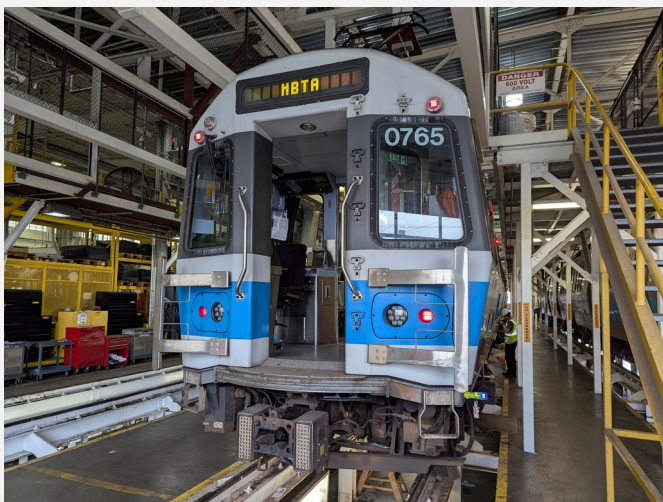
ENSCO, Inc. has used VAMPIRE for many years and has recently obtained the source code to VAMPIRE to continue developments specific to our customer needs



B O S T O

Instrumented Testing

- Test Vehicles for each line tested in both empty and loaded conditions





Instrumented Testing

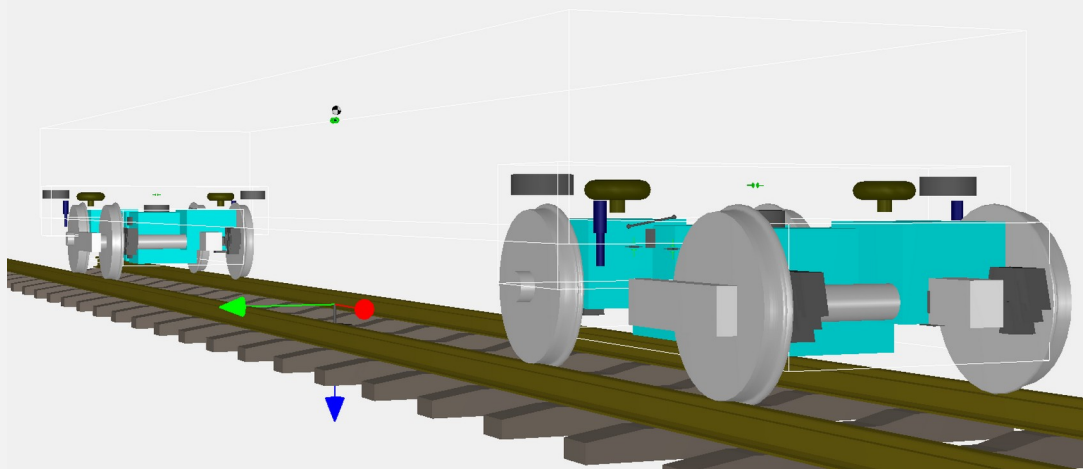


- Laser distance sensors were installed to measure suspension travel
- Accelerometers were installed on axle boxes, truck frames, bolsters and car bodies to measure acceleration responses
- String potentiometers installed to measure truck rotation as the vehicle traveled through curves

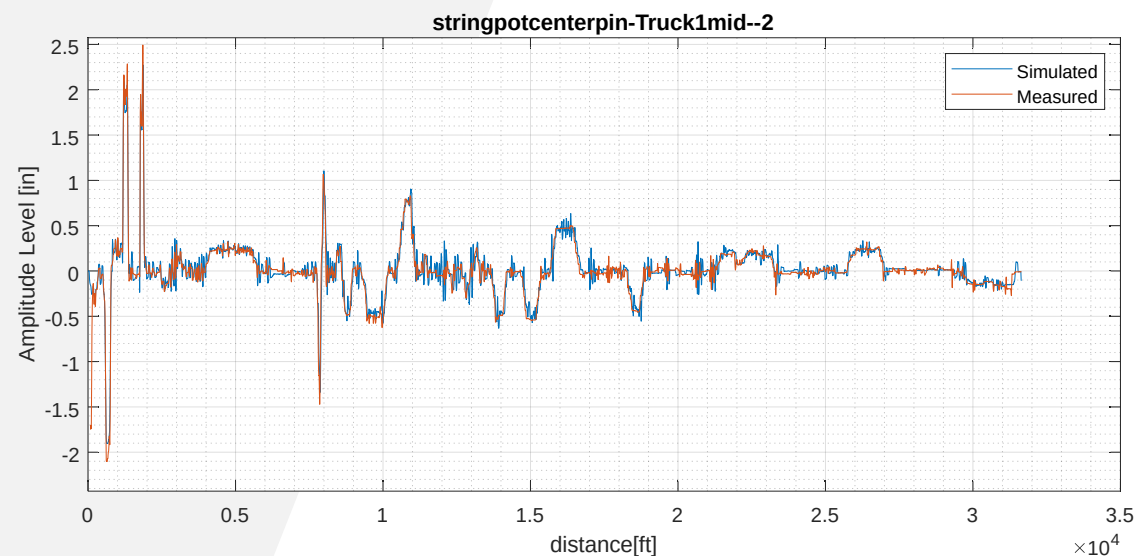
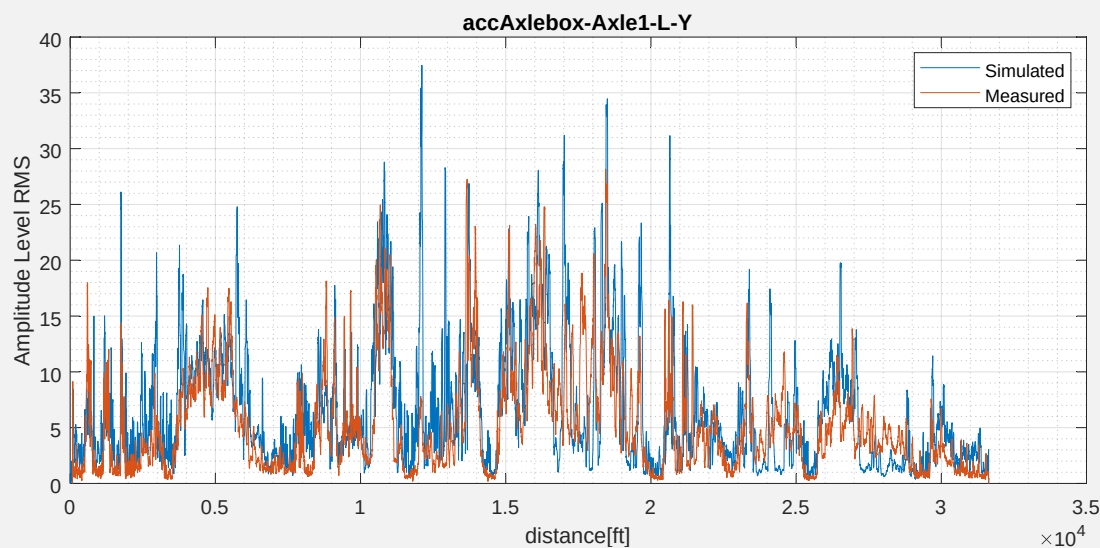




Model Validation



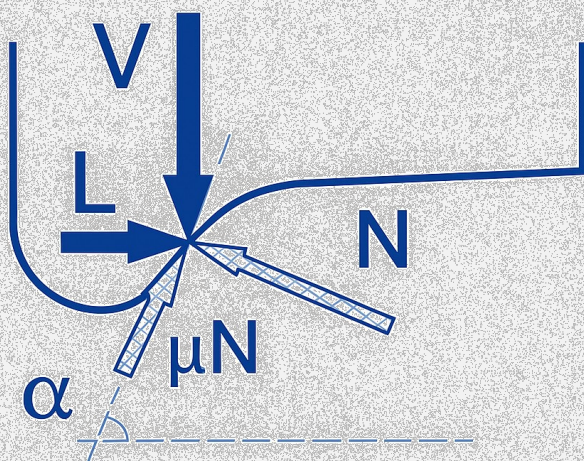
- The models reproduce the measured response envelope, principal elevated-response regions, and overall amplitude range reasonably well.
- The simulated response tends to overpredict at peaks making the models generally conservative.





Derailment Risk – Determined by The Interaction of the System

Nadal's Critical Wheel L/V Ratio



$$\frac{L}{V_{\text{crit}}} = \frac{\tan \alpha - \mu}{1 + \mu \tan \alpha}$$

1. Wheel Flange Contact Angle (α)

- a) The steeper (larger) the flange angle, the harder it is for the wheel to climb (if that location is kinematically available). A shallower angle lowers the threshold, making wheel climb more likely at lower L/V ratios.

2. Coefficient of Friction (μ)

- a) Higher friction increases the wheel's ability to "grip" the rail and climb, which lowers the critical L/V ratio. Lower friction (e.g., wet or lubricated rail) reduces climbing ability, raising the critical limit.

3. Wheel/Rail Contact Conditions

- a) Profile wear: worn wheel or rail shapes can alter the effective flange angle.
- b) Rail inclination: changes how forces resolve at the flange.
- c) Lateral geometry defects (misalignment, gauge issues) can amplify lateral forces.
- d) Wheelset back-to-back spacing can affect contact conditions

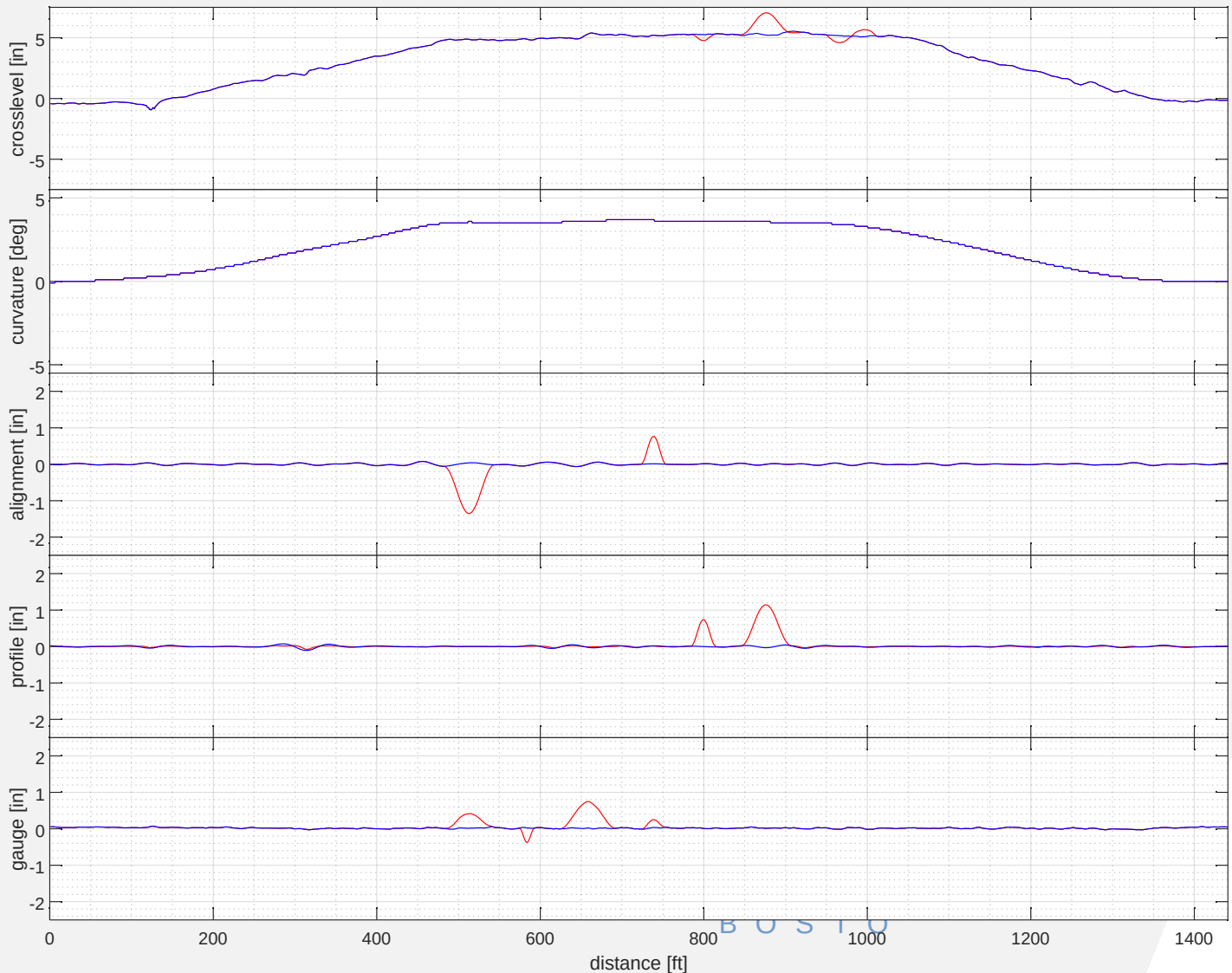
4. Vehicle/Track Dynamics

- a) Suspension stiffness and damping affect how lateral forces build up.
- b) Curvature, crosslevel, lateral alignment, vertical alignment, cant deficiency, and hunting oscillations all influence the applied L/V.
- c) High speeds may amplify dynamic forces, reducing effective safety margins.

Introduced Track Defects: Alignment, Gauge, Profile, All at Their Proposed Thresholds.



Green Line, D Line Eastbound, Chainage 12 432-11 082



Safety Indicator	Dynamic Value
L/V	0.61

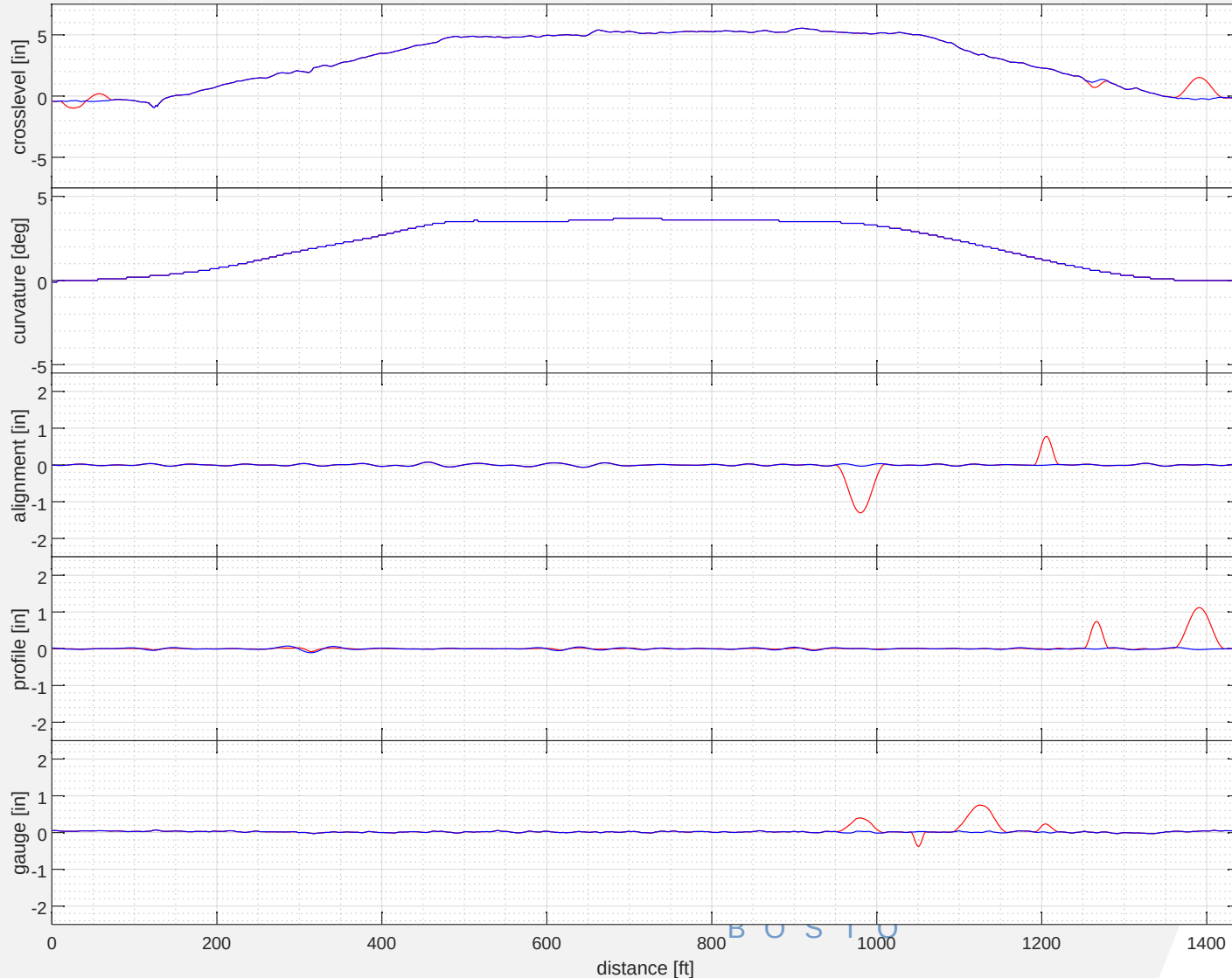
Nadal's Critical Wheel L/V Ratio

$$\frac{L}{V_{crit}} = \frac{\tan \alpha - \mu}{1 + \mu \tan \alpha}$$



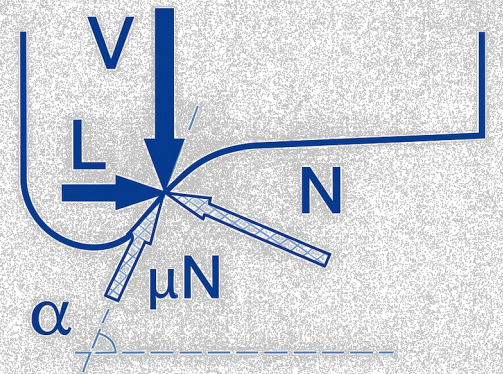
Introduced Track Defects: Alignment, Gauge, Profile, All at Their Proposed Thresholds.

Green Line, D Line Eastbound, Chainage 12 432-11 082



Safety Indicator	Dynamic Value
L/V	0.59

Nadal's Critical Wheel L/V Ratio



$$\frac{L}{V_{\text{crit}}} = \frac{\tan \alpha - \mu}{1 + \mu \tan \alpha}$$



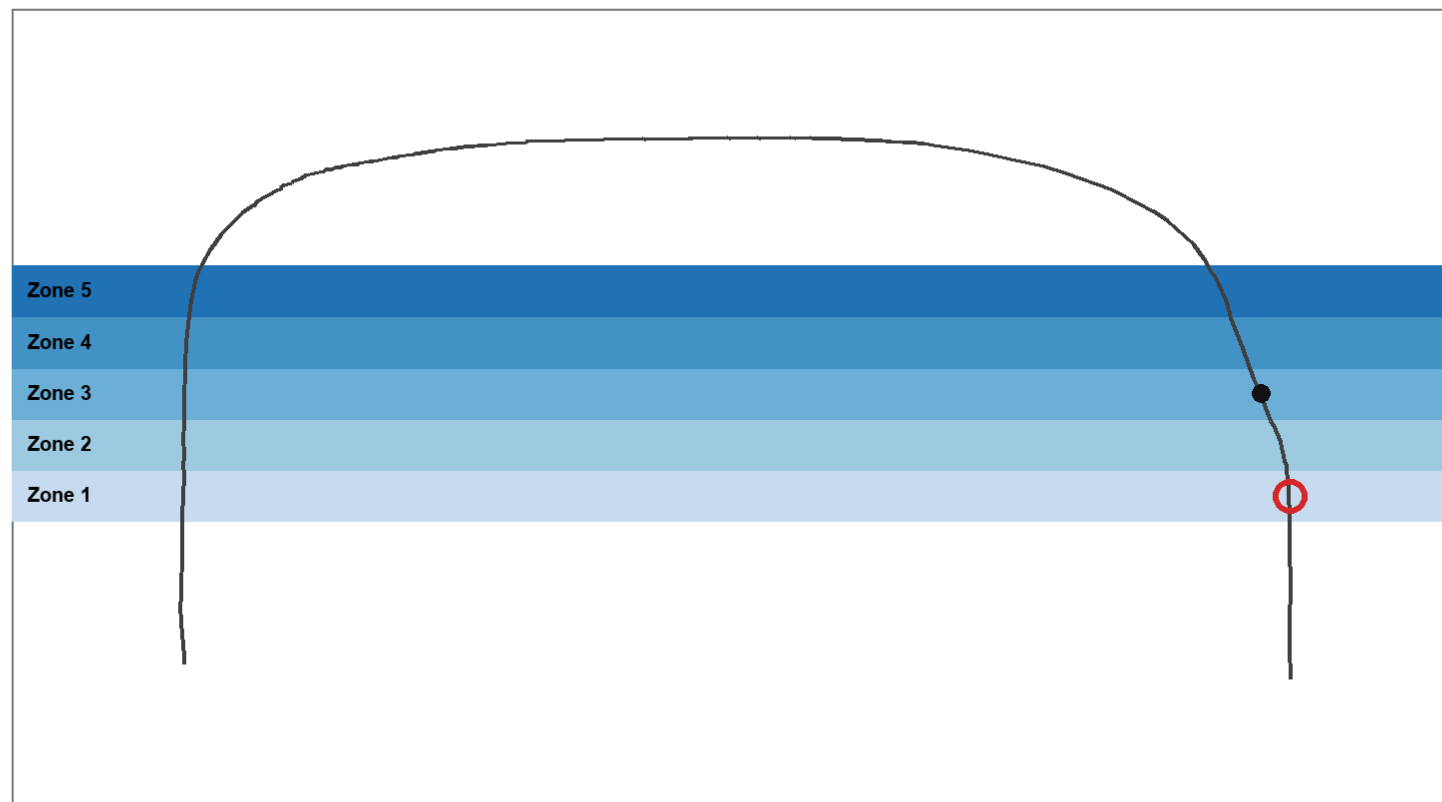
Rail Profile Evaluation Method Review – Further Room for Improvement

- Consistent with established industry practice, the MBTA currently evaluates wheel and rail profiles independently.
- As part of its continuous-improvement efforts, the MBTA is investigating more rigorous system-level methods that consider the interacting wheel–rail geometry and the contact conditions that result.

Current Rail Profile Evaluation Method

Red Line NB Station 56120.97 W high rail — MBTA gauge-face assessment

MBTA selected: Zone 1, rail tangent 1.58 deg from vertical



Gray: measured rail Black dot: 5/8-in gauge point Red circle: MBTA-selected steepest location

- The current MBTA method evaluates the rail gauge face independently of the wheel profile.
- Five vertical zones are established around the standard gauge point (likely path of flange contact migration), and a representative surface angle is calculated within each zone.
- The steepest selected gauge-face angle is treated as the likely available geometric resistance to wheel climb.

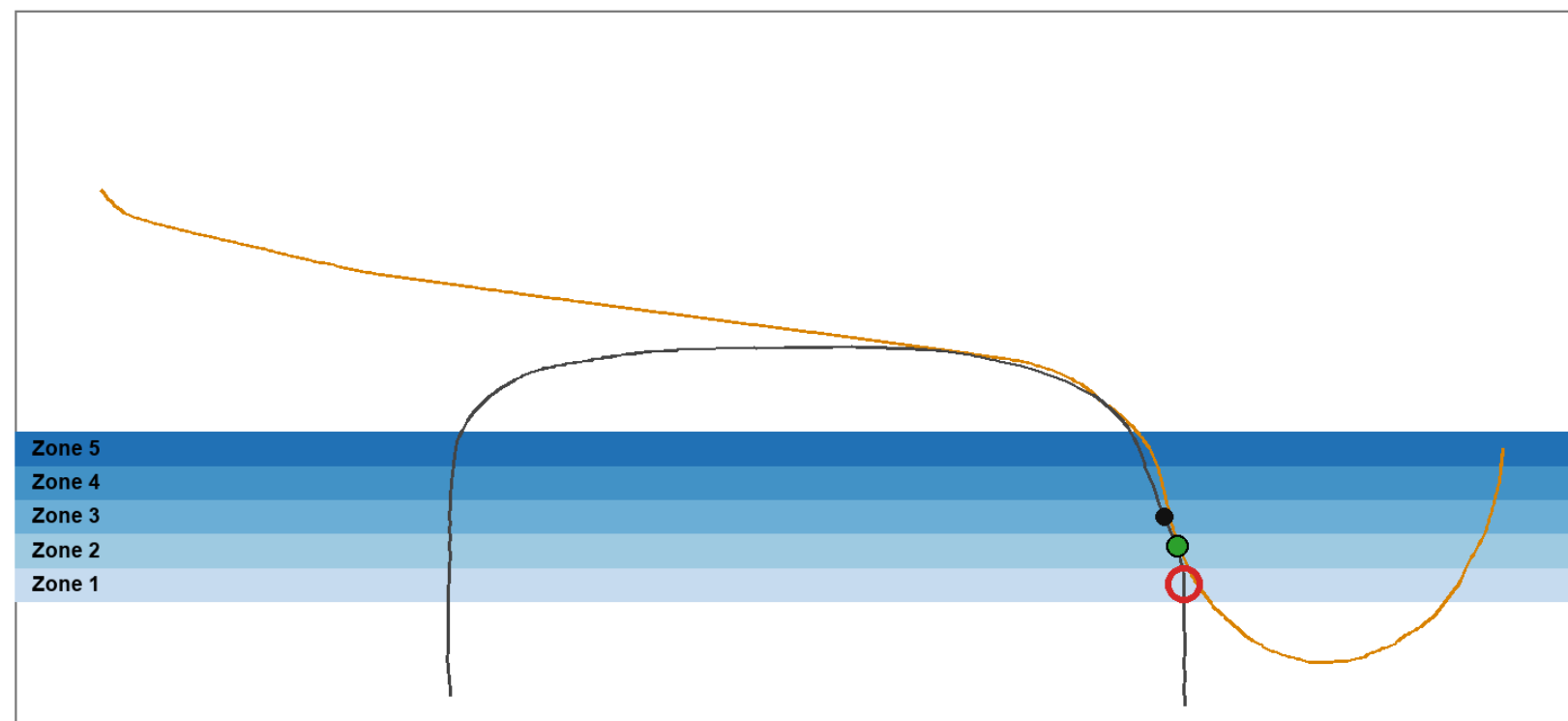


Developmental Wheel/Rail System Evaluation Method

Red Line NB Station 56120.97 W high rail - wheel at incipient climb contact

MBTA selected angle: 1.58 deg from vertical MBTA-angle Nadal L/V: 1.869 at $\mu=0.5$

Wheel/rail contact angle: 23.27 deg from vertical Actual fitted-contact Nadal L/V: 0.844 at $\mu=0.5$



- Quasi-static kinematic calculations of this rail incorporating a measured wheelset, measured geometry, and the opposite rail profile (Wheel/Rail System Evaluation)
- This analysis shows that the identified steep angle gauge face location using the traditional method does not actually participate in the contact climb path.
- **In this case, the identified steep angle would not be a geometric barrier to wheel climb**



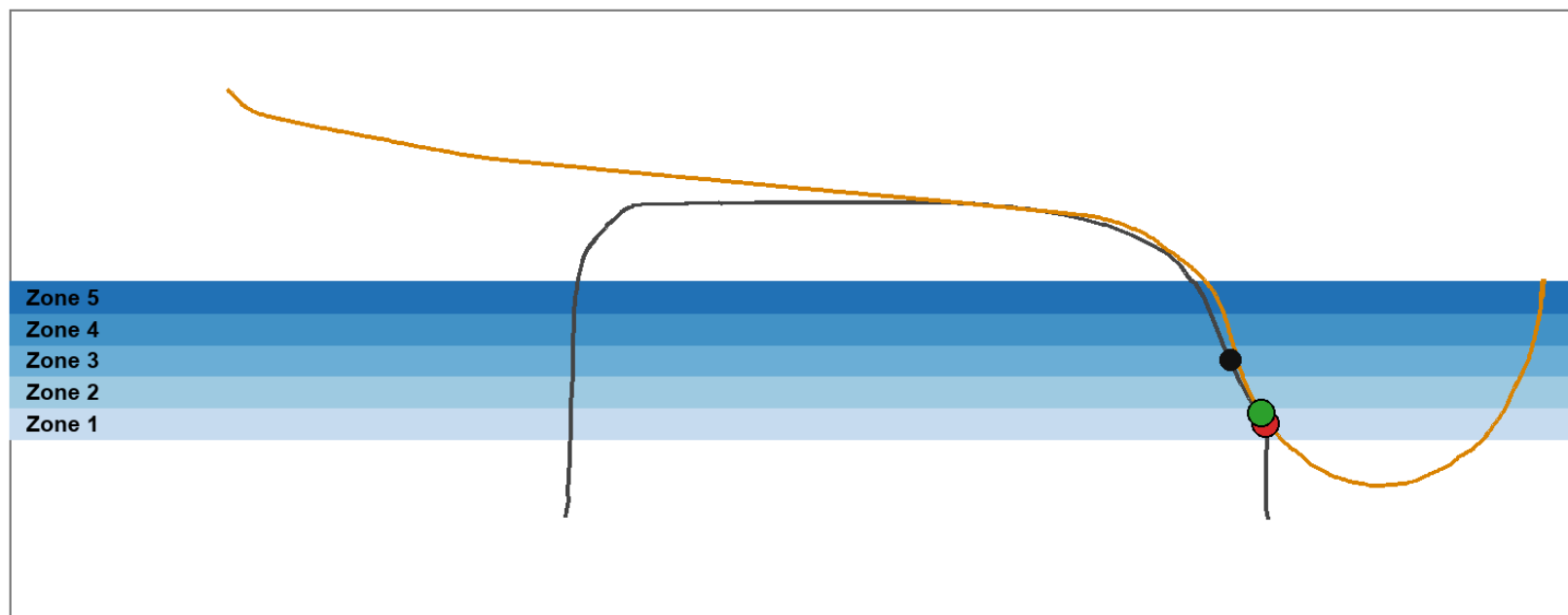
Another Example – System Dictates Contact

Orange Line SB 336+05.14 W high rail - Plot 1 - Initial fitted contact

MBTA Zone 1 point: Nadal L/V 1.101 at $\mu=0.5$

Initial fitted contact: 34.10 deg from vertical, Nadal L/V 0.562

Initial tangent mismatch: 2.68 deg Penetration: 0.000 mm



POINTS SHOWN

- Black dot — 5/8-inch gauge point.
- Red dot — MBTA zone-selected point, 1.03 mm below initial contact.
- Green dot — initial fitted contact; Nadal L/V = 0.562.

WHY THE RAIL PROFILE IS A PRIORITY

- Initial contact is above the geometric barrier identified by the rail only identification method and the initial contact point reveals a relatively low Nadal threshold



The Feasible Path Reaches a Different Gauge-face Barrier

Orange Line SB 336+05.14 W high rail - Plot 2 - Maximum reachable gauge-face contact

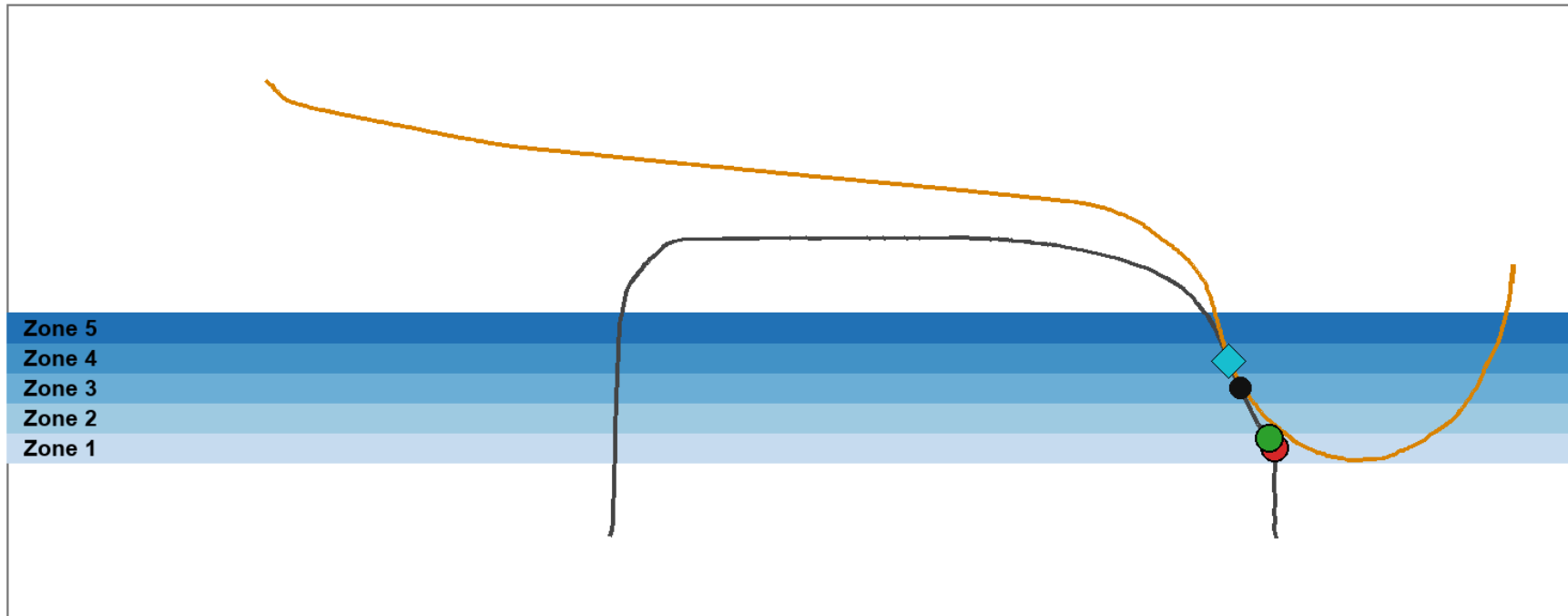
MBTA Zone 1 point: Nadal L/V 1.101 at $\mu=0.5$

Initial fitted contact: 34.10 deg from vertical, Nadal L/V 0.562

Maximum reachable gauge-face contact: 23.47 deg from vertical, Nadal L/V 0.838

Contact rise above initial: 8.19 mm Tangent mismatch: 0.071 deg Penetration: 0.000 mm

- Available on the rail does not necessarily mean available in contact





Maximum Profile Angles Do Not Govern This Contact Path

Orange Line SB 336+05.14 W high rail - Plot 3 - Maximum-angle locations versus reachable contact

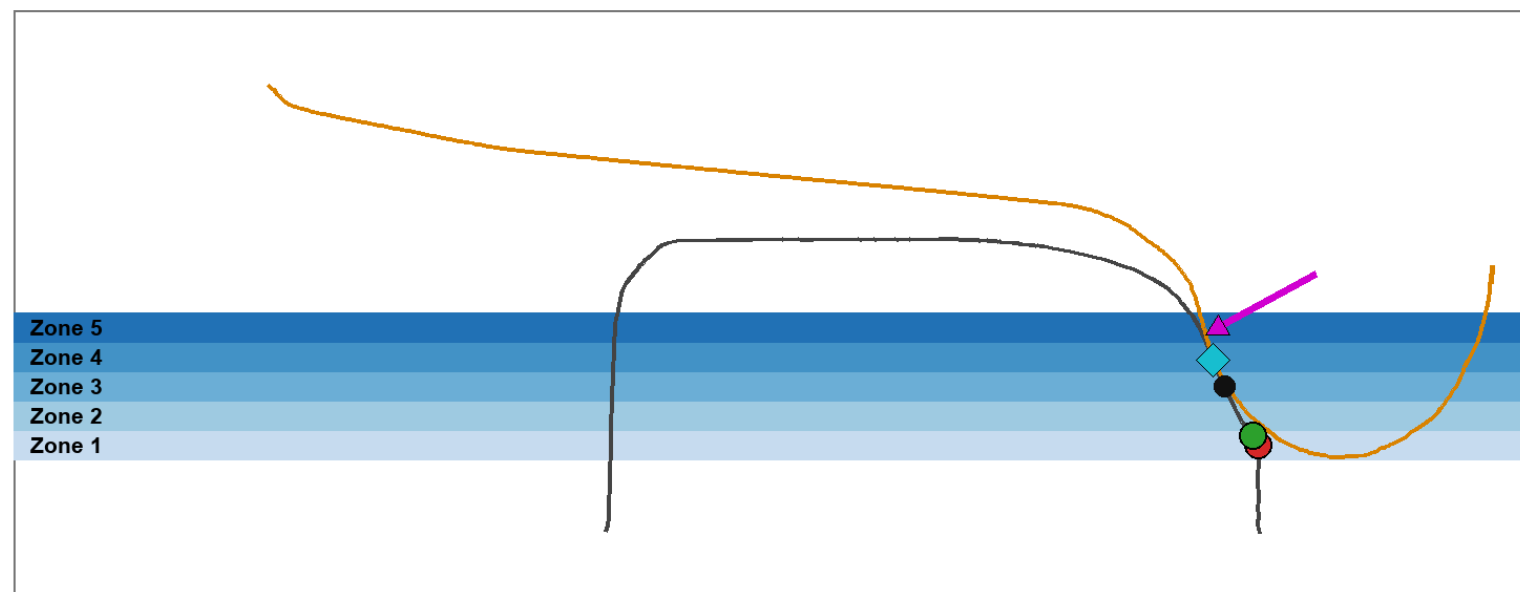
MBTA Zone 1 point: Nadal L/V 1.101 at $\mu=0.5$

Initial fitted contact: 34.10 deg from vertical, Nadal L/V 0.562

Maximum reachable gauge-face contact: 23.47 deg from vertical, Nadal L/V 0.838

Contact rise above initial: 8.19 mm Tangent mismatch: 0.071 deg Penetration: 0.000 mm

Magenta arrow: global maximum flange-angle location; it is not engaged in this feasible pose.



- At smooth, nonpenetrating wheel–rail contact, the wheel and rail must share a common tangent and common normal.
- Either surface may become the geometrically limiting surface: the available wheelset kinematics determine which mutually compatible wheel and rail locations can actually form common-tangent contact.
- **The realized contact angle is a property of the interacting wheel–rail system—not necessarily of either profile considered independently.**



Measured Profiles Create a System-wide Screening Layer



Measured rail-profile network and selected Orange Line corridor

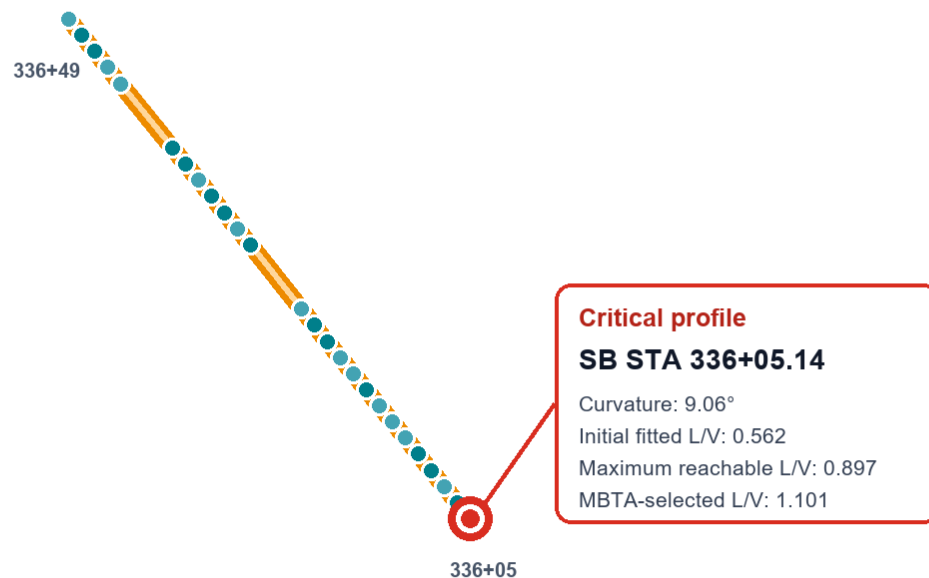
Five-line coverage represented in the rail-profile screening dataset



Focused Analysis of a Single Corridor

Orange Line SB 336+05–336+49 — individual profile locations

44-ft high-rail corridor through a 6.78°–9.06° curve



Dots represent individual measured high-rail profiles; red ring identifies the selected critical profile.



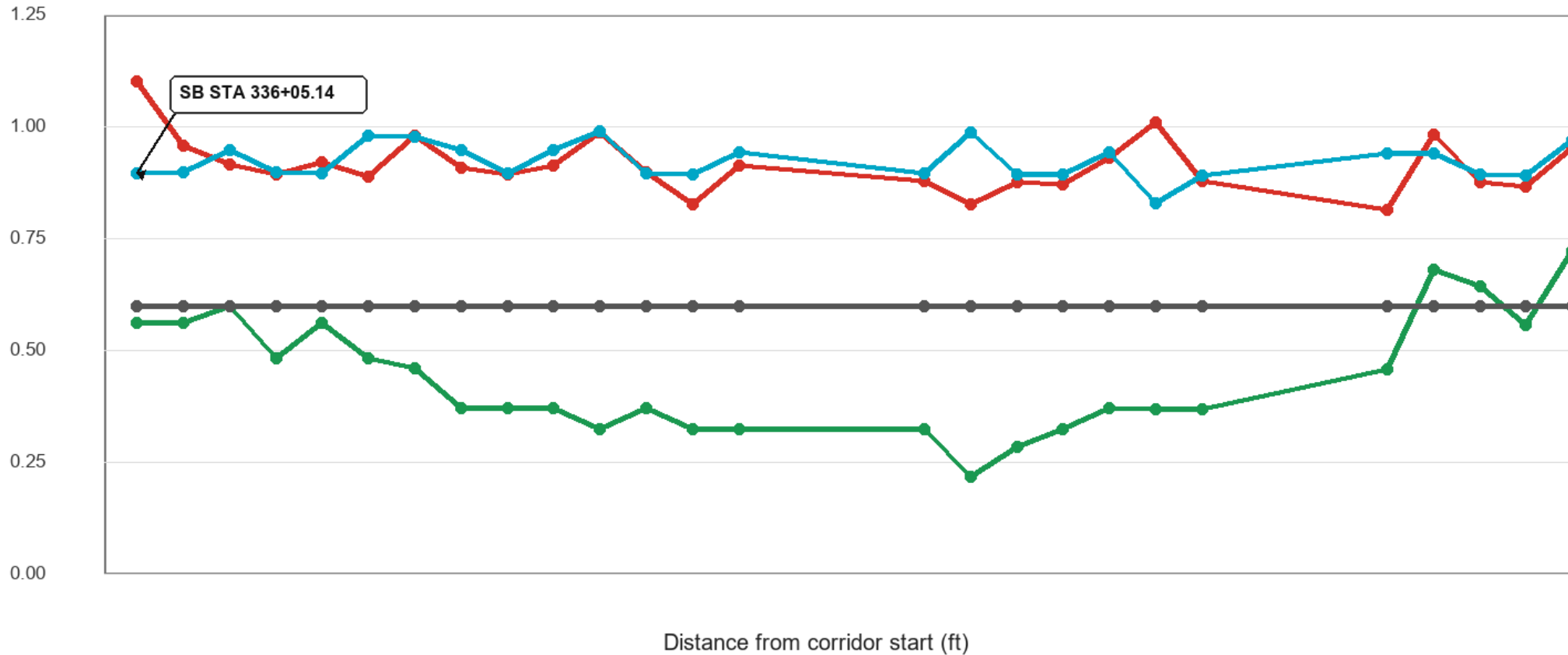
Corridor Traces Reveal Whether Low Thresholds Persist



Orange Line SB 336+05--336+49 - connected-path Nadal screening

— MBTA selected threshold — Initial fitted-contact threshold — Maximum reachable connected-path threshold — 0.600 reference

Nadal L/V at $\mu = 0.5$



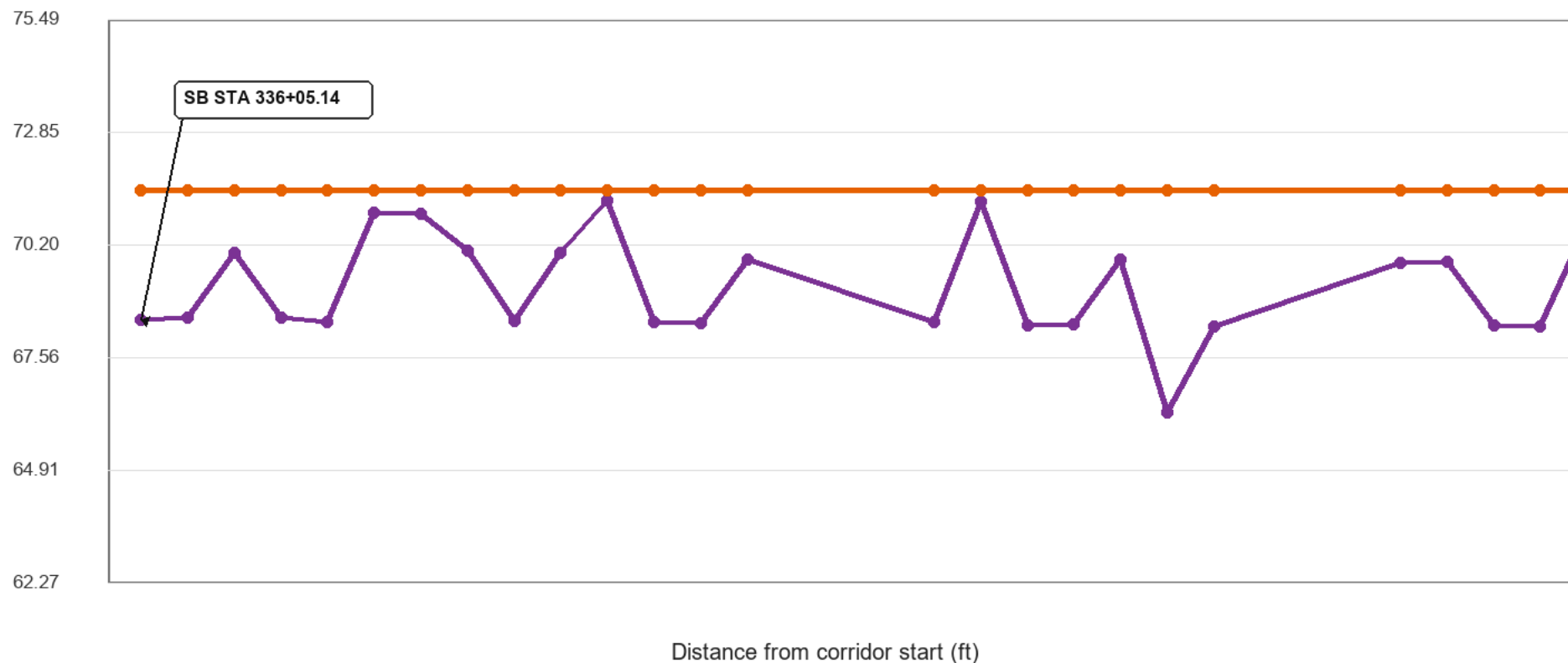
Robust search: 5 deg; STA 336+12.28 and 336+13.71 validated with 7 deg transitions

Maximum Flange Angle Does Not Necessarily Engage in Contact

Orange Line SB 336+05--336+49 - reachable versus global flange angle

— Maximum reachable engaged flange angle — Wheel global maximum flange angle

Flange tangent angle from horizontal (deg)



➤ Available on the wheel does not necessarily mean available in contact





Proposed Maintenance Screening Criteria – Currently under Study

Prioritize locations where the wheel–rail system exhibits:

- A **low incipient-contact Nadal threshold**, indicating greater susceptibility to initiating flange climb.
- A **relatively low maximum reachable connected-path threshold**, indicating limited geometric resistance as contact migrates upward.
- A **large climb distance between the incipient contact and maximum-resistance points**, providing a longer migration path before the strongest reachable geometric barrier is encountered.
- A **persistent pattern across consecutive profiles**, indicating a sustained corridor condition rather than an isolated measurement or profile irregularity.
- An **MBTA-selected point below the incipient contact point**, suggesting that the conventionally selected gauge-face location may not participate in the predicted upward contact path.



Prioritizing Maintenance

- Maintenance priority increases where both the incipient-climb threshold and the maximum reachable connected-path threshold are low, the strongest reachable geometric resistance occurs well above the initial contact point, and the condition persists across consecutive rail profiles.

Developmental Screening Hierarchy (Currently under Investigation)

- **Highest priority:** Low initial and maximum-reachable thresholds, appreciable climb distance, and continuity across multiple sequential profiles.
- **Elevated priority:** Low initial threshold with limited improvement along the reachable contact path.
- **Monitor:** Low initial threshold followed promptly by a substantially stronger reachable geometric barrier.
- **Lower priority:** Isolated low result not repeated in adjacent profiles

A single wheel profile provides only one realization of wheel–rail interaction. **A representative population of wheelsets—including nominal, moderately worn, and limiting profiles—should be evaluated** against the rail network to determine how frequently unfavorable contact conditions occur.



Potential Use for Fleet-level Evaluation of Wheels and Rails

Evaluate a representative wheel-profile population—not a single wheel—against the measured rail network. The frequency and distribution of unfavorable combinations can help determine whether rail reprofiling, wheel reprofiling, or a coordinated treatment provides the most efficient risk reduction.

Useful statistical outputs being considered

- Percentage of wheel profiles producing a priority condition at each rail location
- Distribution and lower percentile of incipient and maximum-reachable Nadal thresholds
- Consecutive rail length over which unfavorable results persist
- Wheel-profile families repeatedly associated with low thresholds
- Expected improvement from simulated wheel or rail reprofiling
- Number of unfavorable wheel–rail combinations eliminated per maintenance dollar

The objective is not merely to identify the “worst” wheel or rail profile, but to select the maintenance action that improves the greatest number of wheel–rail combinations across the operating system.



Kinematic Screening Establishes Maintenance Priority— Not Dynamic Outcome

WHAT THE SCREEN CAN INDICATE

- The initial fitted-contact Nadal threshold indicates relative susceptibility to initiating quasi-static flange climb.
- The maximum reachable connected-path threshold represents the strongest geometric resistance found along the calculated contact path.
- The climb distance between these points indicates how far contact may migrate before reaching that resistance.
- Persistence across consecutive profiles indicates a corridor-scale condition rather than an isolated measurement.

WHAT THE SCREEN CANNOT DETERMINE BY ITSELF

- Whether the operating wheel–rail forces exceed either calculated threshold.
- Whether a dynamically moving wheel will be stopped by the maximum-resistance point.
- Whether every calculated contact state is reached during an actual vehicle passage.
- Whether a flagged profile is unsafe or will cause a derailment.

MAINTENANCE APPLICATION

Prioritize locations having low initial and maximum-reachable thresholds, appreciable climb distance, and persistence across adjacent profiles. Evaluate a representative population of wheel profiles, and use vehicle-dynamic analysis or testing where warranted.



QUESTIONS?